Opening phase as a boundary interaction

A verified 46,173-letter superpermutation on eight symbols

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The reported improvement retains all sixteen closed trails of Jakub Halfar's selected instance of Jay Pantone's construction: four large trails arising from four walks through 27 source loops each, and twelve trails completing the remaining source loops. No trail is split or deleted, and no *designated* permutation window within a trail is discarded. Only the order of the trails and their cyclic opening positions change. The selection of source loops and the search program are Halfar's; the experiment reported here is a boundary reassembly, not a new all-order construction.

Definitions and conventions

A *superpermutation of order* n is a finite word containing all $n!$ permutations of its alphabet as contiguous substrings. An *n-window* is any substring of length n. It is *valid* if its n symbols are distinct and hence form a permutation, and *dirty* otherwise. A *repeated permutation occurrence* is a valid window beyond the first occurrence of its permutation.

We work with an alphabet of n distinct symbols and put $h = n - 3$. For the computation, $n = 8$, $h = 5$, and the alphabet is $\{0, 1, 2, 3, 4, 5, 6, 7\}$. A *cyclic spelling* is a word read around a circle; its positions are understood modulo its length. The *source loops* and *closed trails* refer to the cyclic components and their assembled walks in the selected construction. A *trail opening* cuts a cyclic spelling and extends it just far enough to preserve every designated permutation window on that trail.

1. Phase determines two literal boundaries

For general $n \geq 4$, let trail i have cyclic spelling length R_i. An admissible opening α selects a cyclic starting position for the first designated n-window. Let $g_{i,\alpha}$ be the forward distance (in letters) from the last designated window's start to the first one's start, measured across the cut. In coordinates starting at the first window, the last starts at position $R_i - g_{i,\alpha}$. Thus the linear piece containing both complete windows has length

$$\ell_{i,\alpha} = (R_i - g_{i,\alpha}) + n = R_i + h + a_i(\alpha), \qquad a_i(\alpha) := 3 - g_{i,\alpha}.$$

The number $a_i(\alpha)$ is the *opening adjustment* relative to the baseline $R_i + h$. For the openings in this experiment, a *weight-w opening* means one with gap $g = w$. Consequently, weight-3 openings have $a = 0$, and weight-2 openings have $a = 1$.

Write $P_{i,\alpha}$ for the resulting opened piece. Its starting and ending h-letter words are

$$S_{i,\alpha} = \mathrm{prefix}_h(P_{i,\alpha}), \qquad E_{i,\alpha} = \mathrm{suffix}_h(P_{i,\alpha}).$$

Here *phase* denotes the complete cyclic start position, not merely its residue modulo h (the *coarse clock phase*). Distinct openings with the same coarse phase can expose different S and E.

2. One boundary-cost matrix; coupled opening states

Let $\mathcal{B}_n$ be the set of ordered h-letter words with distinct symbols from the alphabet. Since $h = n - 3$,

$$|\mathcal{B}_n| = n(n-1)\cdots(n-h+1) = \frac{n!}{3!}, \qquad |\mathcal{B}_8| = 6\,720.$$

For $E, S \in \mathcal{B}_n$, define the longest suffix-to-prefix overlap and the resulting boundary cost:

$$\begin{aligned} \mathrm{ov}(E,S) &= \max\bigl\{k \in \{0,\dots,h\} : \mathrm{suffix}_k(E) = \mathrm{prefix}_k(S)\bigr\}, \\ D[E,S] &= h - \mathrm{ov}(E,S). \end{aligned}$$

The length-zero prefix and suffix are both the empty word. The matrix $D \in \mathbb{R}^{|\mathcal{B}_n|\times|\mathcal{B}_n|}$ is *directed*: in general, $D[E,S] \neq D[S,E]$. It can be evaluated entry by entry without allocating a dense array.

For a boundary word b, let $\mathbf{e}_b$ be its *one-hot vector*, with one entry equal to 1 at the index of b in $\mathcal{B}_n$ and all other entries equal to 0. Thus an actual join from E to S costs

$$\mathbf{e}_E^T D\, \mathbf{e}_S = D[E,S].$$

Let $\mathcal{T} = \{0,1,\dots,15\}$ be the set of trail IDs, matching the zero-based saved plan. Suppose trail $i \in \mathcal{T}$ has m_i admissible openings, labeled $\alpha = 1,\dots,m_i$. Define the matrices whose columns are its possible boundary vectors, and its vector of opening adjustments:

$$\begin{aligned} \mathsf{E}_i &= [\,\mathbf{e}_{E_{i,1}} \ \cdots\ \mathbf{e}_{E_{i,m_i}}\,], \\ \mathsf{S}_i &= [\,\mathbf{e}_{S_{i,1}} \ \cdots\ \mathbf{e}_{S_{i,m_i}}\,], \\ \mathbf{a}_i &= (a_i(1),\dots,a_i(m_i))^T. \end{aligned}$$

The complete table of interactions between opening choices of distinct trails i, j is

$$\boxed{C_{ij} = \mathsf{E}_i^T D\, \mathsf{S}_j}, \qquad C_{ij}[\alpha,\beta] = D[E_{i,\alpha}, S_{j,\beta}].$$

An *opening-choice vector* $x_i \in \{0,1\}^{m_i}$ has exactly one nonzero entry: $\mathbf{1}^T x_i = 1$. The nonzero entry selects a single opening of trail i, fixing *both* ends at once. Let $\sigma : \{1,\dots,16\} \to \mathcal{T}$ be a bijection; the sequence $\sigma(1),\dots,\sigma(16)$ specifies trail order. The opening and joining totals are

$$A = \sum_{i\in\mathcal{T}} \mathbf{a}_i^T x_i, \qquad J = \sum_{k=1}^{15} x_{\sigma(k)}^T C_{\sigma(k),\sigma(k+1)} x_{\sigma(k+1)}.$$

In this assembly rule, adjacent pieces are joined using the maximum overlap of their exposed h-letter boundaries (at most h letters). Each piece has length $R_i + h + \mathbf{a}_i^T x_i$; the join of consecutive pieces removes $h - D[E,S]$ repeated boundary letters. Therefore

$$\begin{aligned} L &= \sum_{i\in\mathcal{T}} \bigl(R_i + h + \mathbf{a}_i^T x_i\bigr) - \sum_{k=1}^{15} \bigl(h - x_{\sigma(k)}^T C_{\sigma(k),\sigma(k+1)} x_{\sigma(k+1)}\bigr) \\ &= \boxed{h + \sum_{i\in\mathcal{T}} R_i + A + J}. \end{aligned}$$

This is an exact identity for the *specified opening-and-joining procedure*, not a claim of globally optimal overlap for arbitrary words. Optimizing incoming and outgoing endpoints independently would generally choose two different openings of the same trail and thus describe an unrealizable assembly.

3. An expanded matrix from the new word

The three weight-2 openings appear consecutively in trail order $14, 15, 13$. Trail IDs are zero-based identifiers in the saved plan; offsets are positions in the corresponding cyclic spellings.

trail	offset	g	S	E
14	6538	2	21750	17503
15	8616	2	17503	75032
13	3415	2	03254	32541

Let $t_1 = 14$, $t_2 = 15$, and $t_3 = 13$ denote the trails in the order shown. We now define two 3×3 matrices. The *overlap matrix* O records how many letters are shared by each possible directed pair of the selected trail boundaries:

$$\boxed{O_{ij} := \mathrm{ov}(E_{t_i}, S_{t_j}) = \max\{k \in \{0, 1, \ldots, 5\} : \ \mathrm{suffix}_k(E_{t_i}) = \mathrm{prefix}_k(S_{t_j})\}.}$$

An overlap must match at the *end* of E_{t_i} and the *beginning* of S_{t_j}; a matching substring in the middle does not count. To compute an entry, test $k = 5, 4, \ldots, 1$ and retain the first match, or assign zero if there is none.

The matching boundary strings for all nine ordered pairs are given below; ε denotes the empty word (an overlap of length zero).

	$S_{14} = \texttt{21750}$	$S_{15} = \texttt{17503}$	$S_{13} = \texttt{03254}$
$E_{14} = \texttt{17503}$	ε	17503	03
$E_{15} = \texttt{75032}$	2	ε	032
$E_{13} = \texttt{32541}$	ε	1	ε

Taking the lengths of these strings yields

$$O = \begin{pmatrix} 0 & 5 & 2 \\ 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix}.$$

The *joining-cost matrix* M records the number of additional boundary letters that must be written when joining the same ordered pairs. Since every boundary has length $h = 5$, define

$$\boxed{M_{ij} := 5 - O_{ij} = D[E_{t_i}, S_{t_j}].}$$

Thus M is the 3×3 table extracted from the universal boundary-cost matrix D using rows E_{14}, E_{15}, E_{13} and columns S_{14}, S_{15}, S_{13}. In matrix notation,

$$M = 5\mathbf{1}\mathbf{1}^T - O = \begin{pmatrix} 5 & \boxed{0} & 3 \\ 4 & 5 & \boxed{2} \\ 5 & 4 & 5 \end{pmatrix}, \qquad \mathbf{1} = (1, 1, 1)^T,$$

where $\mathbf{1}\mathbf{1}^T$ is the 3×3 all-ones matrix. For instance, $E_{14} = S_{15} = \texttt{17503}$ gives $O_{1,2} = 5$ and $M_{1,2} = 0$; the three-letter match 032 between $E_{15} = \texttt{75032}$ and $S_{13} = \texttt{03254}$ gives $O_{2,3} = 3$ and $M_{2,3} = 2$.

The entries are directed: $M_{1,2} = 0$ whereas $M_{2,1} = 4$. Diagonal entries are shown for completeness, though the assembly never revisits a trail.

The consecutive joins actually selected are

$$\mathbf{u}_1^T M \mathbf{u}_2 = 0, \qquad \mathbf{u}_2^T M \mathbf{u}_3 = 2,$$

where $\mathbf{u}_1 = (1, 0, 0)^T$, $\mathbf{u}_2 = (0, 1, 0)^T$, and $\mathbf{u}_3 = (0, 0, 1)^T$ index the displayed rows and columns. The first join has the complete five-letter equality $E_{14} = S_{15} = 17503$; the second uses the three-letter equality $\mathrm{suffix}_3(75032) = 032 = \mathrm{prefix}_3(03254)$, so it adds two boundary letters. These are genuine entries for the reported openings; the total gain must be counted across the complete sixteen-trail order.

4. Three extra opening letters save four joining letters

The fixed total cyclic spelling length reported by the assembly plan is

$$\sum_{i \in \mathcal{T}} R_i = 46\,144 = (8! + 7! + 6!) + 64.$$

The extra 64 is part of the fixed trail data, not something derived by this boundary-cost argument. Hence $h + \sum_i R_i = 46\,149$, and the length accounting is

	Halfar reference	new assembly
$h + \sum_i R_i$	46 149	46 149
A	0	3
J	25	21
L	46,174	46,173

For the new plan, the fifteen directed boundary costs are, in order,

$$J = 1 + 1 + 3 + 1 + 1 + 2 + 2 + 0 + 2 + 1 + 1 + 1 + 3 + 1 + 1 = 21.$$

Consequently,

$$\boxed{\Delta L = \Delta A + \Delta J = 3 + (21 - 25) = -1.}$$

Three more letters are needed to open the selected trails, but four fewer letters are needed to join the sixteen pieces. The net saving is collective; it cannot be assigned independently to each changed opening.

5. Coverage, occupation, and scope

For a linear word W, let $q_W(\pi)$ count its contiguous occurrences of permutation π, and define the *occupied support*

$$\operatorname{supp}(W) = \{\pi \in S_n : q_W(\pi) > 0\}.$$

Here S_n denotes the symmetric group, viewed as the set of all permutations of the alphabet. In this note, *occupation-preserving* means that every permutation remains in this support; it does *not* mean that all multiplicities $q_W(\pi)$ are unchanged. All designated permutation windows are preserved inside their complete opened pieces, and joining on equal substrings preserves those pieces. Additional permutation windows spanning a join may change.

Independent sliding-window scans of the supplied reference and improved words give $8! = 40\,320$ distinct permutations in each. The new word has 56 repeated permutation occurrences, compared with 57 in the reference, and both have 5,790 dirty windows. In particular, $46{,}173 - 8 + 1 = 40\,320 + 56 + 5\,790$, as required.

This is an occupation-support-preserving change of boundary phase. The matrix formulation explains the coupled costs; Halfar's randomized `trailsearch` supplies the discrete choices reported in the plan. The argument does not establish minimality or a general analytic selection rule. A discrete Fourier transform diagonalizes the cyclic-shift operator of an internal clock, but that fact alone does not diagonalize the directed boundary matrix D.

Symmetry remark. For every simultaneous relabeling of the alphabet $\pi \in S_n$, the boundary matrix obeys

$$D[\pi(E), \pi(S)] = D[E, S].$$

Thus D commutes with the natural S_n-action on boundary words. This may allow a representation-theoretic block decomposition, but does not on its own give a scalar Fourier diagonalization or solve the opening selection problem.

Reproducibility. Search and attribution: `../PROBE27.md`.

Sixteen-trail assembly plan: `../n8-46173.plan`.

Improved word: `../n8-46173.txt`. A single optional final newline is excluded from length and coverage checks.